LP-2 Practical Viva Preparation DFS and BFS Traversal

This PDF contains important viva preparation material for the LP-2 practical: Depth First Search (DFS) and Breadth First Search (BFS) using Graph Traversal. It includes concepts, algorithms, code explanation, functions used, complexity analysis, and important viva questions with answers.

1. Important Concepts

- **Graph:** A graph is a non-linear data structure consisting of vertices and edges.
- **Vertex:** A node in the graph.
- **Edge:** A connection between two vertices.
- **Undirected Graph:** Edges can be traversed in both directions.
- **Adjacency List:** Stores neighbors of every node using lists.
- **DFS:** Traverses graph depth-wise using recursion or stack.
- **BFS:** Traverses graph level-wise using queue.
- **Visited Array:** Prevents revisiting nodes and avoids infinite loops.

2. Algorithms

DFS Algorithm:
1. Mark starting node as visited.
2. Print/store the node.
3. Visit all unvisited neighbors recursively.
4. Repeat until all reachable nodes are visited.

BFS Algorithm:
1. Insert starting node into queue.
2. Mark it as visited.
3. Remove node from queue.
4. Visit all unvisited neighbors and add them to queue.
5. Repeat until queue becomes empty.

3. Functions and Data Structures Used

Function / DS	Purpose
ArrayList	Stores adjacency list dynamically
List<List<Integer>>	Graph representation
Stack	Used in iterative DFS
Queue	Used in BFS traversal
LinkedList	Implementation of Queue

Scanner	Takes user input
boolean vis[]	Tracks visited nodes
push()	Adds element into stack
pop()	Removes top stack element
offer()	Adds element into queue
poll()	Removes element from queue

4. Code Flow Explanation

Step 1: Create adjacency list for graph representation.

Step 2: Take input for nodes and edges.

Step 3: Store edges in both directions for undirected graph: adj.get(u).add(v); adj.get(v).add(u); **Step 4:** Call recursive DFS function.

Step 5: Call BFS traversal function using queue.

Step 6: Print traversal results.

5. Time and Space Complexity

Algorithm	Time Complexity	Space Complexity
DFS	O(V + E)	O(V)
BFS	O(V + E)	O(V)

6. Important Viva Questions and Answers

Q. What is a graph?

Ans: A graph is a data structure consisting of vertices and edges.

Q. What is DFS?

Ans: DFS is a graph traversal algorithm that explores depth-wise.

Q. What is BFS?

Ans: BFS is a graph traversal algorithm that explores level-wise.

Q. Which data structure is used in BFS?

Ans: Queue is used in BFS.

Q. Which data structure is used in DFS?

Ans: Stack or recursion call stack.

Q. Why is visited array used?

Ans: To avoid revisiting nodes and infinite loops.

Q. Difference between DFS and BFS?

Ans: DFS goes depth-wise while BFS goes level-wise.

Q. Which traversal gives shortest path?

Ans: BFS gives shortest path in unweighted graph.

Q. What is adjacency list?

Ans: A graph representation storing neighbors of each node.

Q. Why adjacency list is preferred?

Ans: It uses less memory for sparse graphs.

End of Viva Preparation Notes